

Дано: $\Delta P_{\text{от}} = \Delta P_a$, $\frac{T_{\text{от}}}{T} = 0,4$, $\frac{V_{\text{кр}}}{V} = 0,09$, $\frac{\Delta T_{\text{от}}}{\Delta T_a} = ?$
 $\frac{\Delta P}{\Delta T} = \frac{c_p}{RT} - b + V \Rightarrow \frac{\Delta T_{\text{от}}}{\Delta T_a} = \frac{\frac{2a}{RT} - b + V}{\frac{2a}{RT} - b + V}$
 найдем $\frac{\Delta T_{\text{от}}}{\Delta T_a} \approx 0,05$.

Второе задание.

№ 19.

$U \in [0; U_0]$, $\langle U \rangle = ?$, $U_{\text{кр}} = ?$, $\Delta U = ?$, $\frac{\Delta U}{\langle U \rangle} = ?$, $\int_0^{U_0} f(U) dU = 1 \Rightarrow$
 $\Rightarrow f(U) = \frac{1}{U_0}$, $\langle U \rangle = \int_0^{U_0} \frac{U}{U_0} dU = \frac{U_0^2}{2U_0} = \frac{U_0}{2}$,
 $\langle U^2 \rangle = \int_0^{U_0} \frac{U^2}{U_0} dU = \frac{U_0^3}{3U_0} = \frac{U_0^2}{3} \Rightarrow U_{\text{кр}} = \frac{U_0}{\sqrt{3}}$,
 $\Delta U = \sqrt{\langle U^2 \rangle - \langle U \rangle^2} = \sqrt{\frac{U_0^2}{3} - \frac{U_0^2}{4}} = \frac{U_0}{2\sqrt{3}}$, $\frac{\Delta U}{\langle U \rangle} = \frac{\frac{U_0}{2\sqrt{3}}}{\frac{U_0}{2}} = \frac{1}{\sqrt{3}}$.

№ 20.
 $M_2, T = 300 \text{ K}$, $U_{\text{кр}} = ?$, $\langle U \rangle = ?$, $U_{\text{кр}} = ?$, $c_p = ?$.

$$N_{He} = \sqrt{\frac{2kT}{m}} \approx 422 \text{ м/с}; \quad \langle v \rangle = \sqrt{\frac{8kT}{\pi m}} \approx 446 \text{ м/с}; \quad v_{He} = \sqrt{\frac{2kT}{m}} \approx 517 \text{ м/с}; \quad C_{He} = \sqrt{\frac{2kT}{m}} \approx 353 \text{ м/с}.$$

№ 4.52.

$$T = 300 \text{ K}; \quad p = 24 \text{ мм рт. ст.} \quad \frac{dm}{dt} = ? \quad Z = \frac{n \langle v \rangle}{4} \quad n = \frac{p}{kT};$$

$$\langle v \rangle = \sqrt{\frac{8kT}{\pi m}} \Rightarrow Z = \frac{p}{4kT} \sqrt{\frac{8kT}{\pi m}} = \frac{p}{\sqrt{2\pi m kT}} \quad \frac{dm}{dt} = Z \cdot m =$$

$$= p \sqrt{\frac{m}{2\pi kT}} = p \sqrt{\frac{M}{2\pi R T}} \approx 38 \frac{\text{кг}}{\text{м}^2 \cdot \text{с}}; \quad \left(\frac{dm}{dt} \right)_{\text{кг}} = 15 \frac{\text{кг}}{\text{м}^2 \cdot \text{с}}.$$

№ 4.18.

$$N, T, j(t) - ? \quad t_0 - ? \quad U_0 - ?$$

$$j = \frac{dn}{S dt} = \frac{dn}{4\pi R^2 dt}; \quad U = \frac{R}{t} \Rightarrow dU = -\frac{R}{t^2} dt.$$

$$dn = N p(U) dU \Rightarrow j(t) = \frac{N}{4\pi R^2} \cdot 4\pi R^2 \left(\frac{m}{2\pi kT} \right)^{3/2} \exp\left(-\frac{mU^2}{2kT} \right) \frac{dU}{dt} =$$

$$= \frac{NR}{t^2} \left(\frac{m}{2\pi kT} \right)^{3/2} \exp\left(-\frac{mR^2}{2kT t^2} \right)$$

$$\frac{dj}{dt} = 0 \Rightarrow N \left(\frac{m}{2\pi kT} \right)^{3/2} \exp\left(-\frac{mR^2}{2kT t^2} \right) \left(-\frac{4R}{t^3} + \frac{R}{t^4} \frac{2mR^2}{2kT t^2} \right) = 0 \Rightarrow$$

$$\Rightarrow \frac{mR^2}{kT t^2} = 4 \Rightarrow t_0 = \frac{R}{2} \sqrt{\frac{m}{kT}}; \quad U_0 = \frac{R}{t_0} = 2 \sqrt{\frac{kT}{m}}.$$



№ 4.14.

$$T = 1150 \text{ K}; \quad V_1 = 0.2 \text{ В}; \quad V_2 = 0.4 \text{ В}. \quad \alpha - ?$$

Для гальванического элемента: $p(U) dU = \left(\frac{m}{2\pi kT} \right)^{3/2} \exp\left(-\frac{mU^2}{2kT} \right) 2U dU$

$$N_\alpha = N p(U, +\infty) \Rightarrow \alpha = \frac{N_\alpha}{N} = p(U, +\infty). \quad \frac{mU^2}{2} = eV \Rightarrow$$

$$\Rightarrow U = \sqrt{\frac{2eV}{m}} \Rightarrow \alpha = \int_0^\infty p(U) dU = \int_0^\infty \frac{m}{kT} \exp\left(-\frac{mU^2}{2kT} \right) U dU =$$

$$= \frac{m}{kT} \cdot \frac{kT}{m} \int_0^\infty \exp\left(-\frac{mU^2}{2kT} \right) d\left(\frac{mU^2}{2kT} \right) = -\exp\left(-\frac{mU^2}{2kT} \right) \Big|_0^\infty = \exp\left(-\frac{m}{2kT} \cdot \frac{2eV}{m} \right) =$$

$$= \exp\left(-\frac{eV}{kT} \right) \Rightarrow \alpha_1 \approx e^{-2} \approx 0.135; \quad \alpha_2 \approx e^{-4} \approx 0.0183.$$

Дано: $T, m, n.$

найти: $z, m, n.$

$$\Rightarrow z = \int_0^\infty dn.$$

$$= n \sqrt{\frac{m}{2\pi kT}} \frac{2k}{2U}$$

$$S = 10^{-6} \text{ м}^2;$$

$$t = 20^\circ \text{ C}; \quad V =$$

eg. $\frac{dU}{dt} = V.$

$$\Rightarrow \frac{dU}{dt} = V.$$

$$\Rightarrow \frac{d(n-n_0)}{n-n_0} = \frac{U}{S} dt$$

$$\Rightarrow t_{12} = \frac{U}{S}$$

$$\Delta U = \pm 3 \text{ мВ}$$

$$\Rightarrow \alpha = 6$$

$$\Rightarrow \alpha = 49$$

гальванический

$$p(U) = \frac{n}{k}$$

$$\Rightarrow \frac{mU_0^2}{kT}$$

№ 4.20.

Сред. T, m, n . Z -? Число молекул, на расстоянии, находящихся
в единицу времени в объёме $V \in (V; V+dV)$: $dn_1 = n p(V) V dV$

$$\Rightarrow Z = \int_0^\infty dn_1 = \int_0^\infty n p(V) V dV = \int_0^\infty n \sqrt{\frac{m}{2\pi kT}} \exp\left(-\frac{mV^2}{2kT}\right) V dV =$$

$$= n \sqrt{\frac{m}{2\pi kT}} \frac{2kT}{2m} = n \sqrt{\frac{kT}{2\pi m}}.$$

№ 4.24.

$S = 10^{-6} \text{ м}^2$, $p_1 = 10^{-4} \text{ мм рт.ст.}$, $p_2 = 10^{-2} \text{ мм рт.ст.}$, $p_0 = 460 \text{ мм рт.ст.}$, $p_3 = \frac{p_0}{2}$,
 $t = 20^\circ \text{C}$, $V = 1 \text{ л}$. t_{12} ? t_{13} ? В соуде в момент t
 в объёме $Z_1 S = \frac{n_0 \langle V \rangle S}{4}$, в момент $Z_2 S = \frac{n(t) \langle V \rangle S}{4}$. $n = \frac{N}{V} \Rightarrow$

$$\Rightarrow \frac{dN}{dt} = V \frac{dn}{dt} = S \frac{\langle V \rangle}{4} (n_0 - n(t)) \Rightarrow \frac{dn}{n_0 - n} = \frac{S \langle V \rangle}{4V} dt.$$

$$\int_{n_0}^n \frac{d(n - n_0)}{n - n_0} = - \frac{S \langle V \rangle}{4V} \int_0^t dt \Rightarrow \frac{-S \langle V \rangle t}{4V} = \ln \frac{n - n_0}{n_0 - n} = - \ln \frac{p_0 - p_1}{p_0 - p_2} \Rightarrow$$

$$\Rightarrow t_{12} = \frac{4V}{S \langle V \rangle} \ln \frac{p_0 - p_1}{p_0 - p_2} \Rightarrow t_{12} \approx 1,13 \text{ с}; t_{13} = 6 \cdot 10^4 \text{ с} \approx 16,7 \text{ ч}.$$

№ 4.40.

$\Delta V = \pm 3 \text{ мкВ}$, $T_1 = 600 \text{ К}$, $T_2 = 400 \text{ К}$. $\frac{\alpha_1}{\alpha_2}$? $|\Delta V| \ll U_{\text{н.б.}} \Rightarrow$

$$\Rightarrow \alpha = 4\pi U_{\text{н.б.}}^2 \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{mU_{\text{н.б.}}^2}{2kT}\right) \Delta V. U_{\text{н.б.}} = \sqrt{\frac{2kT}{m}} \Rightarrow$$

$$\Rightarrow \alpha = 4\pi \frac{2kT}{m} \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(-\frac{m \cdot 2kT}{2kT \cdot m}\right) \Delta V = \frac{4}{e} \sqrt{\frac{m}{2\pi kT}} \Delta V \Rightarrow \frac{\alpha_1}{\alpha_2} = \sqrt{\frac{T_2}{T_1}} \approx 1,22.$$

№ 4.16.

"Дисперсион" уг. раз. $\alpha = \frac{U_{\text{н.б.}}}{U_{\text{н.б.}}} = ?$ Аналогично № 4.14:

$$p(V) = \frac{mV}{kT} \exp\left(-\frac{mV^2}{2kT}\right). p'(U_{\text{н.б.}}) = 0 \Rightarrow \exp\left(-\frac{mU_{\text{н.б.}}^2}{2kT}\right) \left(\frac{m}{kT} - \frac{mU_{\text{н.б.}}^2}{kT^2}\right) = 0 \Rightarrow$$

$$\Rightarrow \frac{mU_{\text{н.б.}}^2}{kT} = 1 \Rightarrow U_{\text{н.б.}} = \sqrt{\frac{kT}{m}}. U_{\text{н.б.}}^2 = \int_0^\infty V^2 p(V) dV =$$

$$= \int_0^{\infty} V^3 \frac{m}{kT} \exp\left(\frac{-mV^2}{2kT}\right) dV = \frac{m}{kT} \frac{4k^2 T^2}{2m^2} = \frac{2kT}{m} \Rightarrow \langle V^2 \rangle = \frac{2kT}{m} \Rightarrow$$

$$\Rightarrow \alpha = \frac{1}{12}$$

N 4.63.

Given: $E \geq E_n = 1 \text{ eV}$, $W = 10^{-3}$; $P_0 = 10 \text{ amu}$, $\mu = 40 \text{ } \mu\text{m/s}$; $T = 1160 \text{ K}$.

$$\frac{dN}{dt} = ? \quad E = \frac{mV^2}{2} \geq E_n \Rightarrow V_0 = \sqrt{\frac{2E_n}{m}}$$

$$\frac{dN}{dt} = \frac{dN \cdot m}{dt}; \quad dN = N \cdot S \cdot dV; \quad n = \frac{P_0}{kT} \Rightarrow \frac{dN}{dt} = N \cdot S \cdot m \cdot W \cdot n$$

$$\int_0^{\infty} V^3 \exp\left(\frac{-mV^2}{2kT}\right) dV = 4\pi V^2 m W \frac{P_0}{kT} \left(\frac{m}{2\pi kT}\right)^{3/2} \int_0^{\infty} V^3 \exp\left(\frac{-mV^2}{2kT}\right) dV =$$

$$= 4\pi P_0 V^2 \sqrt{\frac{m}{2\pi kT}} \left(\frac{E_n}{kT} + 1\right) \exp\left(\frac{-E_n}{kT}\right) \approx 5,1 \text{ } 1/e$$

N 4.64.

Given: T, m . $V_{KB} = ?$, $\langle V \rangle = ?$. $dN = N \cdot dn = 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(\frac{-mV^2}{2kT}\right) V^2 dV$

$$(dN)' = 0 \Leftrightarrow \exp\left(\frac{-mV_{KB}^2}{2kT}\right) \left(3V_{KB}^2 - V_{KB}^3 \cdot \frac{m \cdot 2V_{KB}}{2kT}\right) = 0 \Rightarrow \frac{mV_{KB}^2}{kT} = 3 \Rightarrow$$

$$\Rightarrow V_{KB} = \sqrt{\frac{3kT}{m}}$$

$$\langle V \rangle = \frac{\int_0^{\infty} V \cdot dN}{\int_0^{\infty} dN} = \frac{\int_0^{\infty} V \cdot 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(\frac{-mV^2}{2kT}\right) V^2 dV}{\int_0^{\infty} 4\pi n \left(\frac{m}{2\pi kT}\right)^{3/2} \exp\left(\frac{-mV^2}{2kT}\right) V^2 dV} =$$

$$= \frac{3}{4\pi \sqrt{\pi}} \cdot 2\pi^2 = \frac{3}{4\sqrt{\pi}} = \frac{3}{4} \sqrt{\frac{2kT}{m}} = \frac{3}{2} \sqrt{\frac{kT}{2m}}$$

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N 21.

$\frac{P_0}{P_1} = 2$, $\frac{P_0}{P_2} = 5$; $h_1 = 5,5 \text{ km}$. $h_2 = ?$. $dN = A \exp\left(\frac{-E_p}{kT}\right) dV$

$$n = \frac{dN}{dV} \Rightarrow n = n_0 \exp\left(\frac{-mgh}{kT}\right) = n_0 \exp\left(\frac{-\mu gh}{RT}\right)$$

$$\Rightarrow \frac{P_1}{P_0} = \frac{P_0 \exp\left(\frac{-\mu gh_1}{RT}\right)}{P_0} = \exp\left(\frac{-\mu gh_1}{RT}\right) \Rightarrow \frac{-\mu g}{RT} = \frac{1}{h_1} \ln \frac{P_1}{P_0} \Rightarrow$$

$$\Rightarrow \ln \frac{P_0}{P_2} = \frac{\mu gh_2}{RT} = \frac{h_2}{h_1} \ln \frac{P_0}{P_1} \Rightarrow h_2 = \frac{\ln \frac{P_0}{P_2}}{\ln \frac{P_0}{P_1}} h_1 \approx 12,77 \text{ km}$$

N 22.

$\frac{N_1}{N_0} \xrightarrow{\Delta E}$ $\Delta E = 6,0 \cdot 10^{-21} \text{ J}$; $T = 250^\circ \text{C}$. $\frac{N_1}{N_0 + N_1} = \frac{\frac{N_1}{N_0}}{1 + \frac{N_1}{N_0}}$. $\frac{N_1}{N_0} = \exp\left(\frac{-\Delta E}{kT}\right) \Rightarrow \frac{N_1}{N_0 + N_1} = 0,303$

8.1.
 Найти массу аммиака в баллоне. $mg = ps \Rightarrow M = \frac{ps}{g} =$
 $= \frac{4 \cdot 10^5 \cdot R^2}{g} \Rightarrow M \approx 5,2 \cdot 10^{18} \text{ кг.}$

4.40.
 $N_1 = \frac{N_0}{2}, T_1 = ? \langle E_{\text{kin}} \rangle = \frac{3}{2} kT, \langle E_{\text{kin}} \rangle = 2kT$ (из 4.24).
 $z = \frac{0.507}{4} \Rightarrow \frac{dN}{dt} = \int \frac{n \langle v \rangle}{4}, \frac{dn}{dt} = \int \frac{n \langle v \rangle}{4V}.$
 $E = \frac{3}{2} kT N = \frac{3}{2} kT n V \Rightarrow \frac{dE}{dt} = \frac{3}{2} kV \frac{d(nT)}{dt} = \frac{3}{2} kV \left(T \frac{dn}{dt} + n \frac{dT}{dt} \right) = \frac{3}{2} kV \frac{dn}{dt} \left(T + n \frac{dT}{dn} \right).$
 $\Rightarrow \int \frac{n \langle v \rangle}{4} \cdot \frac{m \langle v^2 \rangle}{2} = \frac{3}{2} kV \frac{5n \langle v \rangle}{4V} \left(T + n \frac{dT}{dn} \right) \Rightarrow m \langle v^2 \rangle =$
 $= 3k \left(T + n \frac{dT}{dn} \right) \Rightarrow T.K. \frac{m \langle v^2 \rangle}{2} = 2kT \Rightarrow T + n \frac{dT}{dn} = \frac{m \langle v^2 \rangle}{3k} =$
 $\frac{4T}{3} \Rightarrow \frac{n}{dn} = \frac{1}{3} \frac{T}{dT} \Rightarrow \frac{1}{3} \frac{dn}{n} = \frac{dT}{T} \Rightarrow \frac{1}{3} \ln \frac{n_1}{n_0} = \ln \frac{T_1}{T_0} \Rightarrow$
 $\Rightarrow \frac{T_1}{T_0} = \left(\frac{n_1}{n_0} \right)^{1/3} = \left(\frac{N_1}{N_0} \right)^{1/3} = \left(\frac{1}{2} \right)^{1/3} \Rightarrow T_1 = (2)^{1/3} T_0 \Rightarrow T_1 \approx 216,7 \text{ K.}$

4.81.
 $r = 3 \text{ мкм}, \theta = \frac{\pi}{6}, N_0 = 600 \text{ мкм}^{-3}, n_0 = 10^{16} \text{ м}^{-3}, T = 300 \text{ K. } n_g, P = ?$
 j_+ - втекаем; j_- - вытекаем $\Rightarrow j_+ = \frac{dN_+}{dt} = \frac{N_0 \sin \theta}{4} \frac{d\theta}{dt} = \frac{n}{4} S = \frac{n}{4} \pi r^2 \frac{8RT}{\sqrt{\pi \mu}}.$
 $= n_0 S N_0 \sin \theta = \pi r^2 N_0 n_0 \sin \theta. j_- = \frac{m \pi r^2 \sqrt{8RT}}{4 \sqrt{\pi \mu}} = \pi n_0 r^2 N_0 \sin \theta \Rightarrow$
 $\Rightarrow n_g = 4 n_0 N_0 \sin \theta \sqrt{\frac{\pi \mu}{8RT}} = n_0 N_0 \sin \theta \sqrt{\frac{2 \pi \mu}{RT}} \Rightarrow n_g \approx 9,5 \cdot 10^{18} \text{ м}^{-3}$
 $\Rightarrow \frac{dE_+}{dt} = E_+ j_+ = \frac{m N_0^2}{2} \cdot \pi n_0 r^2 N_0 \sin \theta; \frac{dE_-}{dt} = E_- j_- =$
 $= 2kT n_0 N_0 \sin \theta \sqrt{\frac{\pi \mu}{8RT}} \pi r^2 \frac{8RT}{\sqrt{\pi \mu}} = 2kT n_0 N_0 \sin \theta \pi r^2$
 $P = \frac{dE}{dt} = \frac{dE_+}{dt} - \frac{dE_-}{dt} = \pi r^2 n_0 N_0 \sin \theta \left(2kT - \frac{m N_0^2}{2} \right) = 2kT \pi r^2 n_0 N_0 \sin \theta.$
 $\left(1 - \frac{m N_0^2}{4RT} \right) \Rightarrow P = 6,68 \cdot 10^{-5} \text{ Вт.}$

N 8.11.

$g = 9,8 \text{ m/s}^2$; $\frac{N_r}{N_{\text{os}}} = 4$; $T = 200 \text{ K} = \text{const}$. $N_{\text{gr}}?$

$n(z) = n_0 \exp\left(\frac{-\mu g z}{RT}\right) \Rightarrow N = n_0 S \int_0^{\infty} \exp\left(\frac{-\mu g z}{RT}\right) dz$

$\int_0^{\infty} e^{-ax} dx = \frac{1}{a} \Rightarrow N = \frac{n_0 S RT}{\mu g} \Rightarrow \frac{N_r}{N_{\text{os}}} = \frac{n_r^0 \mu_{\text{os}}}{\mu_r n_{\text{os}}^0} = 4 \Rightarrow n_r^0 = n_{\text{os}}^0$

$\mu_{\text{gr}} = \sqrt{2} \frac{RT}{\mu}$. $\mu = \frac{\mu_{\text{os}} + \mu_r}{2} = 16 \text{ m.u.}$, $\mu = \frac{5/2 R + 5/2 R}{2} = \frac{5}{2} R$

$\Rightarrow N_{\text{gr}} = \sqrt{\frac{3}{2} \cdot \frac{8,31 \cdot 200}{9,016}} \approx 394,4 \text{ m.u.}$



N 8.14. $z_1 = \frac{R}{2}$, $z_2 = \frac{3R}{2}$. $T?$

$n = n_0 \exp\left(\frac{-mgz}{kT}\right)$. $R^2 = r^2 + (R-z)^2 \Rightarrow r^2 = z(2R-z)$

$\Rightarrow dN(z) = n S dz = n_0 \exp\left(\frac{-mgz}{kT}\right) (2Rz - z^2) dz$

$\frac{dN}{dz} = 0 = \exp\left(\frac{-mgz}{kT}\right) (2R - 2z - (2Rz - z^2) \frac{mg}{kT}) \Rightarrow \frac{mg}{kT} (2Rz - z^2) = 2(R-z)$

$\Rightarrow \frac{mg}{kT} = \frac{2(R-z)}{z(2R-z)} \Rightarrow \text{при } z_1 = \frac{R}{2} \Rightarrow \frac{mg}{kT_1} = \frac{4}{3R} \Rightarrow T_1 = \frac{3}{4} \frac{mgR}{k}$

при $z_2 = \frac{3}{2} R \Rightarrow T_2 = -\frac{3}{4} \frac{mgR}{k}$ - невозможна $\Rightarrow T = \frac{3}{4} \frac{mgR}{k}$

N 8.55.

$\mu_F = 22 \cdot 10^3 \text{ m.u.}$, $\rho = 1,1 \text{ g/cm}^3$; $r_0 = 10 \text{ cm}$; $\omega = 10^3 \text{ c}^{-1}$; $T = 295 \text{ K}$. $r_0?$

$m = m_0 - m_g = m_0 \left(1 - \frac{\rho_g}{\rho_0}\right) = \frac{\mu_0}{N_A} \frac{\rho_0 - \rho_g}{\rho_0}$. $F = m \omega^2 r \Rightarrow$

$\Rightarrow F_r = \int_{r_0}^r F dr = \frac{-m \omega^2 r^2}{2} \Rightarrow n(r) = n_0 \exp\left(\frac{m \omega^2 r^2}{2kT}\right)$

$N = \int_0^{r_0} n(r) \cdot 2\pi r dr = \int_0^{r_0} 2\pi r n_0 \exp\left(\frac{m \omega^2 r^2}{2kT}\right) dr = \frac{2\pi n_0 kT}{m \omega^2}$

$\cdot \left(\exp\left(\frac{m \omega^2 r_0^2}{2kT}\right) - 1\right)$. $n = \frac{N}{\pi r_0^2} \Rightarrow \frac{2\pi n_0 kT}{m \omega^2 r_0^2} \left(\exp\left(\frac{m \omega^2 r_0^2}{2kT}\right) - 1\right) =$

$= n_0 \exp\left(\frac{m \omega^2 r_0^2}{2kT}\right) \Rightarrow \frac{m \omega^2 r_0^2}{2kT} = \ln\left(\frac{kT}{m \omega^2 r_0^2} \left(\exp\left(\frac{m \omega^2 r_0^2}{2kT}\right) - 1\right)\right) \Rightarrow$

$$\Rightarrow r_k = \frac{2kT}{m\omega^2} \ln \left(\frac{kT}{m\omega^2 r_0^2} \exp \left(\frac{m\omega^2 r_0^2}{2kT} - 1 \right) \right), \text{ где } m = \frac{M}{N_A} \frac{\rho_s - \rho_0}{\rho_s} \Rightarrow r_k \approx 8,1 \mu\text{m}$$

Дано: $p_0, \mu, T, \alpha, g, \gamma, h^*$?

Условия: $\frac{dN}{N} = dV_0 = \frac{n_0 dV}{N_0}$; $dV = \pi (h \tan \alpha)^2 dh$.

$$= \frac{p_0}{RT} \exp \left(-\frac{\mu g h}{RT} \right) dV \Rightarrow \gamma = \frac{p_0}{RT} \pi \tan^2 \alpha \int_0^h h^2 \exp \left(-\frac{\mu g h}{RT} \right) dh =$$

$$= \frac{p_0 \pi}{RT} \tan^2 \alpha \left(\frac{RT}{\mu g} \right)^3 \int_0^{\frac{\mu g h}{RT}} x^2 e^{-x} dx = 2 \pi p_0 \tan^2 \alpha \left(\frac{RT}{\mu g} \right)^3$$

$$dV \propto h^2 \exp \left(-\frac{\mu g h}{RT} \right) dh \Rightarrow \ln(dV) = 2 \ln h - \frac{\mu g h}{RT} = \text{const} \Rightarrow$$

$$\Rightarrow (dV)' = \frac{2}{h} - \frac{\mu g}{RT} = 0 \Rightarrow h^* = \frac{2RT}{\mu g}$$

✓ 8.44.

$\frac{n^{(35)}(r_0)}{n^{(37)}(r_0)} = \alpha_2 = 3,065$; $V = 5 \cdot 10^4 \text{ cm}^3$; $T = 300 \text{ K}$. α_2 ?

$F = m\omega^2 r$

$$\Rightarrow E_n = \int_r F dr = \frac{-m\omega^2 r^2}{2} = -\frac{\mu V^2(r)}{2N_A} \Rightarrow n(r) = n_0 \exp \left(\frac{\mu V^2(r)}{2RT} \right)$$

$$dN = n(r) dV = n_0 \exp \left(\frac{\mu V^2(r)}{2RT} \right) 2\pi r h dr = n_0 h \pi \exp \left(\frac{\mu V^2(r)}{2RT} \right) \frac{dV^2}{\omega^2} \Rightarrow$$

$$\Rightarrow N = \int_0^{\omega^2} n_0 h \pi \exp \left(\frac{\mu V^2}{2RT} \right) \frac{dV^2}{\omega^2} = \frac{n_0 h \pi 2RT}{\mu \omega^2} \left(\exp \left(\frac{\mu V^2}{2RT} \right) - 1 \right)$$

$$N = \frac{N}{\pi h \omega^2} = \frac{2RT n_0}{\mu r_0 \omega^2} \left(\exp \left(\frac{\mu V^2}{2RT} \right) - 1 \right) \Rightarrow \frac{n_{35}}{n_{37}} = \frac{n_{35}}{n_{37}} \frac{\mu_{35} \exp \left(\frac{\mu_{35} V^2}{2RT} \right) - 1}{\mu_{37} \exp \left(\frac{\mu_{37} V^2}{2RT} \right) - 1}$$

$$\frac{n_{35}(r_0)}{n_{37}(r_0)} = \alpha_2 = \frac{n_{35}}{n_{37}} \frac{\mu_{35} (1 - \exp \left(-\frac{\mu_{35} V^2}{2RT} \right))}{\mu_{37} (1 - \exp \left(-\frac{\mu_{37} V^2}{2RT} \right))} \Rightarrow \alpha_2 \approx 2,96$$

✓ 8.25.

$M = 40 \text{ g/mol}$, $a = 2,5 \text{ cm}$, $\omega = 2 \cdot 10^3 \text{ s}^{-1}$; $T = 300 \text{ K}$. ΔC ?

$F = m\omega^2 r$

$$\Rightarrow E = \frac{-m\omega^2}{2} (r^2 - a^2) \quad \langle E \rangle = \frac{-m\omega^2}{2} \frac{\int_0^a r^2 \exp \left(-\frac{m\omega^2 (r^2 - a^2)}{2kT} \right) dr}{\int_0^a \exp \left(-\frac{m\omega^2 (r^2 - a^2)}{2kT} \right) dr} = \frac{-m\omega^2}{2} \frac{\int_0^a r^2 e^{m\omega^2 r^2 / 2kT} dr - a^2 \int_0^a e^{m\omega^2 r^2 / 2kT} dr}{\int_0^a e^{m\omega^2 r^2 / 2kT} dr}$$

$$\frac{m\omega^2 a^2}{2kT} = 0,02 \ll 1 \Rightarrow \langle E \rangle \approx \frac{-m\omega^2}{2} \frac{\int_0^a r^2 (1 + \gamma r^2) dr}{\int_0^a (1 + \gamma r^2) dr} = \frac{-m\omega^2}{2} \frac{\frac{a^4}{4} + \frac{\gamma a^6}{6}}{\frac{a^2}{2} + \frac{\gamma a^4}{4}} \Rightarrow$$

3

$$\Rightarrow \langle E_{osc} \rangle \approx \frac{m\omega^2}{2} \left(a^2 - \frac{a^2}{2} - \frac{2A^4}{6} \right); \quad \gamma = \frac{m\omega^2}{2kT} \Rightarrow \langle E_{osc} \rangle \approx \frac{m\omega^2 a^2}{4} - \frac{(m\omega^2 a^2)^2}{48kT}$$

$$\Delta C = \frac{d\langle E_{osc} \rangle}{dT} = \frac{(m\omega^2 a^2)^2}{48kT^2} = \frac{(m\omega^2 a^2)^2}{48RT^2} \Rightarrow \Delta C \approx 2.9 \cdot 10^{-4} \frac{J}{K \cdot mol}$$

№ 23.

$$R_{ex}, d = 0.34 \cdot 10^{-8} \text{ m}; \quad L, E_{exc} = \frac{3}{2} kT; \quad E_{gr} = \frac{N^2}{2I}; \quad M^2 = \hbar^2 L(L+1)$$

$$I = mr^2 = m \frac{d^2}{4} \quad E_{exc} = E_{gr} \Rightarrow \frac{3}{2} kT = \frac{M^2}{2I} \Rightarrow$$

$$\Rightarrow T = \frac{4\hbar^2 L(L+1)}{3km d^2} \Rightarrow T \approx 108 \text{ K}$$

№ 24.

$$a_{ex}, \quad \omega = 10^{14} \text{ s}^{-1}; \quad T = ?; \quad C_v = ?$$

$$E_{exc} = \hbar\omega, \quad kT > E_{exc} \Rightarrow T > \frac{\hbar\omega}{k} \Rightarrow$$

$$\Rightarrow T > 425 \text{ K}, \quad C_v = \left(\frac{3}{2} + \frac{2}{2} + \frac{2}{2} \right) R = \frac{7}{2} R$$

№ 25.

$$T_1 = 299 \text{ K}; \quad T_2 = 300 \text{ K}; \quad \Delta E = 10^{-19} \text{ J}; \quad \frac{P_{21}}{P_{12}} = ? \quad V = \text{const} \Rightarrow$$

$$\Rightarrow dE = T dS \Rightarrow dS = \frac{dE}{T}$$

$$+ \left(\frac{\partial S}{\partial E_2} \right) \Delta E_2 = \Delta E \left(\frac{1}{T_1} - \frac{1}{T_2} \right); \quad \Delta S_{12} = \Delta E \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$p \propto G \Rightarrow \Delta p \propto \Delta G \Rightarrow p \propto e^{\frac{\Delta S}{k}} \Rightarrow \frac{P_2}{P_1} = \exp \left(\frac{\Delta S}{k} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) \right) e^S$$

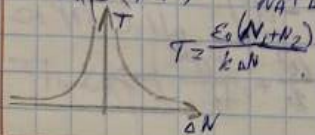
№ 8.58 + 59.

$$\frac{E_2 = E_0}{E_1 = -E_0}; \quad \nu = 1, \text{ мал}; \quad 1 - \text{ам, мал}; \quad N_2 - N_1 = \Delta N, \quad T = ? \quad C_v = ?$$

$$N_1 = N_A W = N_A \frac{\exp \left(-\frac{E_0}{kT} \right)}{\exp \left(-\frac{E_0}{kT} \right) + \exp \left(\frac{E_0}{kT} \right)}; \quad N_2 = N_A \frac{\exp \left(\frac{E_0}{kT} \right)}{\exp \left(-\frac{E_0}{kT} \right) + \exp \left(\frac{E_0}{kT} \right)} \Rightarrow$$

$$\Rightarrow \Delta N = N_A \frac{\exp \left(\frac{E_0}{kT} \right) - \exp \left(-\frac{E_0}{kT} \right)}{\exp \left(-\frac{E_0}{kT} \right) + \exp \left(\frac{E_0}{kT} \right)} = N_A \frac{1 - \exp \left(-\frac{2E_0}{kT} \right)}{1 + \exp \left(\frac{2E_0}{kT} \right)} \Rightarrow$$

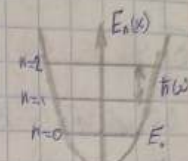
$$\Rightarrow \exp \left(\frac{2E_0}{kT} \right) = \frac{N_A - \Delta N}{N_A + \Delta N} \Rightarrow T = \frac{2E_0}{k \ln \frac{N_A - \Delta N}{N_A + \Delta N}} \quad G \propto \frac{N!}{N_1! N_2!}$$

T.K. log 1-амалнае, $C_v = \frac{3}{2} R + \left(\frac{\partial S}{\partial T} \right)_V$

$$E_0(N_2 - N_1) = E_0 N_A \left(\frac{\exp(-\frac{E_0}{kT}) - \exp(-\frac{E_0}{kT})}{\exp(-\frac{E_0}{kT}) + \exp(-\frac{E_0}{kT})} \right) \Rightarrow \left(\frac{\partial E}{\partial T} \right) = \frac{E_0^2 N_A}{kT^2} \left(1 - \frac{\exp(-\frac{E_0}{kT})}{\exp(-\frac{E_0}{kT}) + \exp(-\frac{E_0}{kT})} \right) = \frac{E_0^2 N_A}{kT^2} \left(1 - \frac{\exp(-\frac{E_0}{kT})}{2 \exp(-\frac{E_0}{kT})} \right) = \frac{E_0^2 N_A}{kT^2} \left(1 - \frac{1}{2} \right) = \frac{E_0^2 N_A}{2kT^2}$$

$$\Rightarrow C_V = \frac{3}{2} R + R \frac{E_0^2}{2kT^2} \quad \sqrt{8.52}$$

$T = 300 \text{ K}; \nu_1 = 10^{13} \text{ Hz}; \nu_2 = 10^{14} \text{ Hz}; \nu_3 = 4.4 \cdot 10^{13} \text{ Hz}$
 $E_n = n E_0 = n h \nu$



Вероятности по Больцману: $P_n = A \exp(-\frac{E_n}{kT})$, $\sum_{n=0}^{\infty} P_n = 1 = A \sum_{n=0}^{\infty} \exp(-\frac{E_n}{kT})$

$$\Rightarrow P_n = \frac{\exp(-\frac{E_n}{kT})}{\sum_{n=0}^{\infty} \exp(-\frac{E_n}{kT})}, \quad \langle E \rangle = \frac{1}{N} \sum_{n=0}^{\infty} N_n E_n = \sum_{n=0}^{\infty} \frac{N_n}{N} E_n = \sum_{n=0}^{\infty} P_n E_n = \frac{\sum_{n=0}^{\infty} n E_0 \exp(-\frac{n E_0}{kT})}{\sum_{n=0}^{\infty} \exp(-\frac{n E_0}{kT})} = \frac{E_0 \sum_{n=0}^{\infty} n \exp(-\frac{n E_0}{kT})}{\sum_{n=0}^{\infty} \exp(-\frac{n E_0}{kT})}$$

$$= \frac{E_0 \exp(-\frac{E_0}{kT})}{(1 - \exp(-\frac{E_0}{kT}))^2} \Rightarrow \langle E \rangle = \frac{E_0 \exp(-\frac{E_0}{kT})}{1 - \exp(-\frac{E_0}{kT})} = \frac{h \nu}{\exp(\frac{h \nu}{kT}) - 1}$$

$\Rightarrow \langle E_1 \rangle \approx 1.64 \cdot 10^{-21} \text{ Дж}; \langle E_2 \rangle \approx 4.35 \cdot 10^{-22} \text{ Дж}$
 $C_k = N_A \frac{d\langle E \rangle}{dT} = N_A \left(\frac{h \nu}{\exp(\frac{h \nu}{kT}) - 1} \right)^2 = R \frac{(\frac{h \nu}{kT})^2 \exp(\frac{h \nu}{kT})}{(\exp(\frac{h \nu}{kT}) - 1)^2} \Rightarrow C_k \approx 0.03 R$

ТЗ.

а5. $S = \frac{4}{2}; E_m = m \mu B; m = -s; (-s+1); \dots; s; \nu = 1 \text{ мал}; B_0 \gg \frac{kT}{\mu}; B_0 \rightarrow 0, T = 1 \text{ К}$
 $\Delta S = ?; Q = ?$ Т.к. E_m уменьшается до 0 ($B_0 \rightarrow 0$)
 $\Delta S = k \ln G$. $\exists 8$ вариантов спина $\Rightarrow \exists 8$ ур. энергии, которые при низком T примерно равновероятны $\Rightarrow$
 $\Rightarrow G = \frac{N!}{(\frac{N}{8})^8}$. Т.к. $\nu = 1 \text{ мал} \Rightarrow N = N_A \Rightarrow \ln G =$
 $= \ln N! - 8 \ln \left(\frac{N}{8} \right) \approx N \ln N - \frac{8 \cdot N}{8} \ln \frac{N}{8} = N \ln 8 \Rightarrow \Delta S = k N \ln 8$
 $Q = T \Delta S \Rightarrow Q = 17.28 \text{ Дж}$
 $\Rightarrow C \approx 17.28 \text{ Дж/К}$

сосуды N_1 N_2 $T_1 = T_2$; $p_1 = p_2$. а). Газы разнородные. б). Газы однородные.

а). $\Delta S = k \ln G = k \ln \frac{(N_1 + N_2)!}{N_1! N_2!} \approx k((N_1 + N_2) \ln(N_1 + N_2) - N_1 \ln N_1 - N_2 \ln N_2) = k(N_1 \ln \frac{N_1 + N_2}{N_1} + N_2 \ln \frac{N_1 + N_2}{N_2})$.

б). т.к. газ в сосуде один и тот же и находится при одинаковых T и p , то $\Delta S = 0$, потому что в действительности расширение газа не происходит (мг. Труден).

T_8 .

NO: $\theta_{gr} \approx 3k$; $\theta_{rot} \approx 2.6 \cdot 10^3 K$; $\epsilon = 0.015 eB$; $T_1 = 50 K$; $T_2 = 300 K$.

$V = 1$ моль. $Q = ?$ $Q = Q_0 + Q_{gen}$. $Q_0 = C_V (T_2 - T_1) = \frac{5}{2} R (T_2 - T_1)$.

$\theta_{gen} = \frac{\epsilon}{k} \approx 144 K \Rightarrow \theta_{gen} \in [T_1; T_2]$. $\langle \epsilon \rangle = \epsilon \frac{\exp(-\frac{\epsilon}{kT})}{1 + \exp(-\frac{\epsilon}{kT})} =$

$= \frac{\epsilon}{1 + \exp(\frac{\epsilon}{kT})}$. $Q_{gen} = N_A (\langle \epsilon_2 \rangle - \langle \epsilon_1 \rangle) = \epsilon N_A \left(\frac{1}{1 + \exp(\frac{\epsilon}{kT_2})} - \frac{1}{1 + \exp(\frac{\epsilon}{kT_1})} \right) \approx$

$\approx 474 \text{ Дж}$; $Q_0 \approx 5200 \text{ Дж} \Rightarrow Q \approx 5,7 \text{ кДж}$.

$\sqrt{8.40}$.

HD: $T_K = 22 K$; $\theta = \frac{\hbar^2}{2Ik} = 64 K$. $C_{gr} = ?$ $\text{для } T_K: \frac{\hbar^2}{2I} l(l+1) = kT$

$\Rightarrow \theta(l(l+1)) = T \Rightarrow l^2 + l + \frac{22}{64} = 0 \Rightarrow l_{1,2} = -\frac{1}{2} \pm 0,74 \Rightarrow l = 0,24$

$\Rightarrow l = \{0; 1; -1\} \Rightarrow g = 3$. $\langle \epsilon \rangle = \frac{g l \epsilon \cdot \exp(-\frac{\epsilon}{kT})}{\exp(-\frac{\epsilon}{kT}) + g_l \exp(-\frac{\epsilon}{kT})} =$

$= \frac{(2l+1) \frac{\hbar^2}{2I} l(l+1) \exp(-\frac{\hbar^2 l(l+1)}{2IkT})}{1 + (2l+1) \exp(-\frac{\hbar^2 l(l+1)}{2IkT})} \approx \frac{3 \hbar^2 \exp(-\frac{\hbar^2}{2IkT})}{I(1 + 3 \exp(-\frac{\hbar^2}{2IkT}))}$. $C_{gr} = \frac{d\langle \epsilon \rangle}{dT} =$

$= \frac{3k \hbar^2 \exp(-\frac{\hbar^2}{2IkT})}{(IkT)^2 (1 + 3 \exp(-\frac{\hbar^2}{2IkT}))} \approx \frac{3k \left(\frac{2\theta}{T}\right) \exp(-\frac{2\theta}{T})}{1 + 3 \exp(-\frac{2\theta}{T})} \approx 0,3 k$.

$\sqrt{8.61}$.

$F_2: V = 3,42 \cdot 10^{13} \text{ Дж}$; $T = 300 K$; $l = \{0; 1\}$. $J = ?$ $J = \frac{C_p}{C_V} = \frac{3,5R + C_{gr}}{2,5R + C_{gr}}$

$$\epsilon_{\text{max}} = h\nu \Rightarrow \langle \epsilon \rangle = \epsilon_{\text{max}} \frac{\exp(-\frac{\epsilon_{\text{max}}}{kT})}{\exp(-\frac{\epsilon_{\text{max}}}{kT}) + \exp(-\frac{\epsilon_{\text{max}}}{kT})} \quad C_{\text{max}} = N_A \frac{d\langle \epsilon \rangle}{dT} =$$

$$= N_A \frac{h^2 \nu^2}{kT^2} \frac{\exp(-\frac{h\nu}{kT})}{(1 + \exp(-\frac{h\nu}{kT}))^2} = R \left(\frac{h\nu}{kT} \right)^2 \exp(-\frac{h\nu}{kT}) \approx 0,125 R \Rightarrow$$

$$\Rightarrow \gamma \approx \frac{3,5 + 0,125}{2,5 + 0,125} \approx 1,38.$$

✓ 9.46.

Дано: $N; m = (-1; 0; 1); \epsilon_0 = 0; \epsilon_{\pm 1} = \epsilon; 0 < \epsilon \ll kT$. $S = ?$

$$\langle \epsilon \rangle = \frac{\sum \epsilon_i e^{-\frac{\epsilon_i}{kT}}}{\sum e^{-\frac{\epsilon_i}{kT}}} = \frac{2\epsilon \exp(-\frac{\epsilon}{kT})}{1 + 2\exp(-\frac{\epsilon}{kT})} \approx \frac{2\epsilon(1 - \frac{\epsilon}{kT})}{3 - 2\frac{\epsilon}{kT}} \approx \frac{1}{3} (2\epsilon - 2\frac{\epsilon^2}{kT}) (1 + \frac{2}{3}\frac{\epsilon}{kT}) \approx$$

$$\approx \frac{2\epsilon}{3} - \frac{2}{3}\frac{\epsilon^2}{kT} + \frac{4}{9}\frac{\epsilon^2}{kT} = \frac{2}{3}\epsilon - \frac{2}{9}\frac{\epsilon^2}{kT}. \quad E = N\langle \epsilon \rangle; dE = TdS \Rightarrow$$

$$\Rightarrow \frac{dE}{dT} = T \frac{dS}{dT} = \frac{2}{9} \frac{N\epsilon^2}{kT^2} \Rightarrow dS = \frac{2}{9} \frac{N\epsilon^2}{kT^3} dT \Rightarrow S = -\frac{1}{9} \frac{N\epsilon^2}{kT^2} + \text{const.}$$

✓ 26.

$m = 12; l = 1 \text{ м}; T \approx 300 \text{ К}$. $\langle \Delta r^2 \rangle = ?$ $\langle \Delta r^2 \rangle = mgl$.

$(1 - \cos \alpha) \approx mgl \left(\frac{\alpha^2}{2} \right) = 2 \cdot \frac{kT}{2} \Rightarrow \langle \alpha^2 \rangle = \frac{2kT}{mgl}$.

$\Delta r \approx l\alpha \Rightarrow \sqrt{\langle \Delta r^2 \rangle} = \sqrt{\frac{2kTl}{mg}} \rightarrow \sqrt{\langle \Delta r^2 \rangle} \approx 0,9 \text{ мм.}$

4

✓ 24.

$V = 1 \text{ мм}^3$. $\epsilon = ?$ $\epsilon \approx \frac{1}{N}$. $pV = NkT \Rightarrow N = \frac{pV}{kT} \Rightarrow \epsilon \approx \sqrt{\frac{kT}{pV}} \approx 2 \cdot 10^{-4}$.

✓ 28.

$k = 1 \frac{\text{м}}{\text{с}}; T \approx 300 \text{ К}$. $\sqrt{\langle x^2 \rangle} = ?$ $\frac{k\langle x^2 \rangle}{2} = \frac{kT}{2}$ (с.м. об.) $\Rightarrow \sqrt{\langle x^2 \rangle} = \sqrt{\frac{kT}{k}} \approx 6,4 \cdot 10^{-11} \text{ м.}$

✓ 9.6.

$\alpha = \frac{\sqrt{\langle \Delta r^2 \rangle}}{\sqrt{\langle n \rangle}} = \frac{1}{\sqrt{\langle n \rangle}} = 10^{-6} \Rightarrow \langle n \rangle = \frac{1}{\alpha^2} = 10^{12} \text{ см}^{-3}; V = \frac{\langle n \rangle kT}{p} =$

$= \frac{kT}{\alpha^2 p} \Rightarrow V \approx 3,4 \cdot 10^{-14} \text{ м}^3$.

✓ 9.8.

$N = 10^{-3} N_A = 6 \cdot 10^{20} \text{ см}^{-3}$. $\frac{\Delta T}{T} = ?$ (мощность микроволн). $pV = NkT \Rightarrow$

$\Rightarrow \Delta T = \frac{p \Delta V}{kN} \Rightarrow \frac{\Delta T}{T} = \frac{\Delta V}{V}$. ϵ - диэлектрическая проницаемость; σ - м. вытгб

поверхн. натяжения; x - линейные размеры зерна $\Rightarrow$
 $\Rightarrow \frac{\langle x^2 \rangle}{2} = \frac{kT}{2} \Rightarrow \langle x^2 \rangle = \frac{kT}{2}$; $F = p\sigma = \left(\frac{\partial p}{\partial N}\right) \sigma x = \left(\frac{\partial p}{\partial N}\right) \sigma^2 x = -\frac{3}{2} \sigma^2 x \Rightarrow$
 $\Pi = - \int F dx = - \int \left(\frac{\partial p}{\partial N}\right) \sigma^2 x dx = - \left(\frac{\partial p}{\partial N}\right) \sigma^2 \frac{x^2}{2} = - \frac{3}{2} \frac{\sigma^2}{2} \Rightarrow \frac{3}{2} = - \sigma^2 \left(\frac{\partial p}{\partial N}\right)$
 $\langle \Delta V^2 \rangle = \sigma^2 \langle x^2 \rangle = \sigma^2 \frac{kT}{2} = -kT \left(\frac{\partial V}{\partial p}\right) \left(\frac{\partial p}{\partial N}\right) = \frac{-kT}{p^2} = -\frac{V}{p} \Rightarrow$
 $\Rightarrow \langle \Delta V^2 \rangle = kT \cdot \frac{V}{p} \Rightarrow \frac{\Delta T}{T} = \frac{\Delta V}{V} = \frac{\sqrt{\langle \Delta V^2 \rangle}}{V} = \sqrt{\frac{kT}{pV}} = \sqrt{\frac{kT}{NkT}} = \frac{1}{\sqrt{N}} \approx 4 \cdot 10^{-8}$

№ 9.28.

Кольцо из: $r = 0,01 \text{ м}$; $T = 300 \text{ К}$; $\beta_T = 5,9 \cdot 10^{-6} \text{ амм}^{-1}$. $\varepsilon = \frac{\sqrt{\langle \Delta V^2 \rangle}}{V}$?

$\frac{3}{2} \frac{\langle \Delta V^2 \rangle}{2} = \frac{kT}{2}$, $F = p\sigma = \left(\frac{\partial p}{\partial N}\right) \sigma \Delta r = \left(\frac{\partial p}{\partial N}\right) \sigma^2 \Delta r = -\frac{3}{2} \sigma^2 \Delta r \Rightarrow$
 $\Rightarrow \Pi = - \int F dr = \left(\frac{\partial p}{\partial N}\right) \sigma^2 \frac{\Delta r^2}{2} = \frac{3}{2} \frac{\Delta r^2}{2} \Rightarrow \frac{3}{2} = \frac{\sigma^2}{\beta_T V} \Rightarrow$
 $\Rightarrow \langle \Delta V^2 \rangle = \frac{kT \beta_T V}{\sigma^2} \Rightarrow \langle \Delta V^2 \rangle = \sigma^2 x^2 = kT \beta_T V \Rightarrow \varepsilon = \frac{\sqrt{\langle \Delta V^2 \rangle}}{V} = \sqrt{\frac{kT \beta_T}{V}}$

№ 9.33.

$\Delta T = 0,1 \text{ К}$; $\lambda = 1 \text{ мкм} = 10^{-6} \text{ м}$; $V_0 = 25 \text{ л}$; $T_0 = 300 \text{ К}$; $p_0 = 10^5 \text{ Па}$. $\frac{W_A}{W_0}$?

(W_A - прирост; W_0 - работа). $W \propto L^3 \Rightarrow \frac{W_A}{W_0} = L^3$. $V \approx \text{const} \Rightarrow N_2: N$

$\Rightarrow \Delta E = V C_V \Delta T + (V_0 - V) C_V \Delta T_0 = 0 \Rightarrow \Delta T_0 = \frac{-V \Delta T}{V_0 - V}$

$\Delta S = V C_V \ln \frac{\Delta T}{T_0} + (V_0 - V) C_V \ln \frac{\Delta T_0}{T_0}$. Т.к. $\Delta T \ll T_0$ и $V \ll V_0$, то

пр.т. получаем $\Delta S \approx -\frac{1}{2} V C_V \left(\frac{\Delta T}{T_0}\right)^2 = -\frac{1}{2} \frac{p_0 V}{R T_0} \cdot \frac{3}{2} R \left(\frac{\Delta T}{T_0}\right)^2 =$
 $= -\frac{3}{4} \frac{p_0 V \Delta T^2}{T_0^3} \Rightarrow \frac{W_A}{W_0} = \exp\left(-\frac{3}{4} \frac{p_0 V \Delta T^2}{k T_0^3}\right) \approx e^{-2} \approx 0,135$

Мембраны;

№ 9.40.

$S \sim 10^{-4} \text{ м}^2$; $p \sim 1 \text{ атм}$; $\tau \sim 10^{-3} \text{ с}$. $\frac{\sqrt{\langle \Delta p^2 \rangle}}{p}$? За пр. т на

мембрану полагаем $N = \frac{1}{4} n \langle v \rangle \tau S$ расм. $\langle v \rangle = \sqrt{\frac{3kT}{m}}$

$n = \frac{p}{kT} \Rightarrow \frac{\sqrt{\langle \Delta p^2 \rangle}}{p} = \frac{\sqrt{\langle \Delta N^2 \rangle}}{N} \approx \frac{1}{\sqrt{N}} = \sqrt{\frac{k}{2Sp} \sqrt{\frac{2\pi m}{R}}} \sim 10^{-8}$

$\rho, T, \sigma; \theta = 0$. $\frac{K \Delta T}{h}$? ~ 9.31 .
 $\Delta p = \rho g \Delta h \Rightarrow F = \rho g \Delta h S \Rightarrow$
 $\Rightarrow \Pi = \frac{\rho g \Delta h S}{2} = \frac{kT}{2} \Rightarrow \langle \Delta h \rangle = \frac{kT}{\rho g}$. $h = \frac{20 \text{ as } 0}{\rho g} =$
 $\frac{25}{\rho g} \Rightarrow \frac{\sqrt{kT}}{h} = \sqrt{\frac{kT}{\rho g}} \cdot \frac{\rho g}{20} = \sqrt{\frac{kT \rho g}{40000}}$.

~ 9.35
 $\frac{K}{\langle \Pi \rangle}$? $\text{Quas } V = 1 \text{ mull} - ?$ $\Pi = \frac{Kx^2}{2} \cdot \frac{K}{\Delta T} = \gamma \Rightarrow$
 $\Rightarrow \langle \Pi \rangle = \frac{\int_0^\infty x^2 \exp(-\gamma x^2) dx}{\int_0^\infty \exp(-\gamma x^2) dx} \cdot \frac{K}{2} = \frac{K}{2} \cdot \frac{\frac{1}{2} \sqrt{\frac{\pi}{\gamma}} \cdot \frac{1}{2}}{\frac{1}{2} \sqrt{\frac{\pi}{\gamma}}} = \frac{K}{4\gamma} = \frac{K \Delta T}{4K} = \frac{\Delta T}{4}$;
 $\langle \Pi^2 \rangle = \frac{\int_0^\infty x^4 \exp(-\gamma x^2) dx}{\int_0^\infty \exp(-\gamma x^2) dx} \cdot \frac{K^2}{4} = \frac{K^2}{4} \cdot \frac{\frac{1}{2} \cdot \frac{3}{4} \sqrt{\frac{\pi}{\gamma}} \cdot \frac{1}{2}}{\frac{1}{2} \sqrt{\frac{\pi}{\gamma}}} = \frac{3K^2}{16\gamma^2} = \frac{3}{4} K^2 T^2$
 $\langle \Delta \Pi^2 \rangle = \langle (\Pi - \langle \Pi \rangle)^2 \rangle = \langle \Pi^2 - 2\Pi\langle \Pi \rangle + \langle \Pi \rangle^2 \rangle = \langle \Pi^2 \rangle - \langle \Pi \rangle^2 = \left(\frac{3}{4} - \frac{1}{4}\right) K^2 T^2 =$
 $= \frac{K^2 T^2}{2} \Rightarrow \frac{\sqrt{\langle \Delta \Pi^2 \rangle}}{\langle \Pi \rangle} = \frac{\sqrt{\frac{K^2 T^2}{2}}}{\frac{\Delta T}{4}} = \sqrt{2}$. $\text{Diel Lennel: } \langle \Pi \rangle = \frac{N_A \cdot K T}{2}$;
 $\langle \Delta \Pi^2 \rangle = N_A \frac{K^2 T^2}{2} \Rightarrow \frac{\sqrt{\langle \Delta \Pi^2 \rangle}}{\langle \Pi \rangle} = \sqrt{\frac{2}{N_A}} \approx 1,8 \cdot 10^{-12}$.

N_2 : $N, Y, \eta = 18 \cdot 10^{-6} \text{ Pa} \cdot \text{s}$. K ? D ? d ? ~ 29
 $\approx 6,6 \cdot 10^{-5} \frac{\text{m}^2}{\text{s}}$; $K = \frac{\rho R}{\mu} = \frac{5}{2} \frac{\eta R}{\mu} \approx 0,05 \frac{\text{Pa}}{\mu \cdot \text{K}}$; $D = \frac{1}{3} \lambda \langle v \rangle =$
 $= \frac{\sqrt{\frac{3RT}{2\pi M}}}{3} \Rightarrow d = \sqrt{\frac{3RT}{2\pi \mu K}} \approx 3,6 \cdot 10^{-10} \text{ m}$.

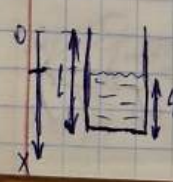
$\sim 30^\circ$
 T_1, T_2 $h = 23 \text{ mm}$; $\Delta T = 30^\circ \text{C}$;
 $K = 2,3 \cdot 10^{-2} \frac{\text{Pa}}{\mu \cdot \text{K}}$. q ? $q = K \frac{dT}{dx} \Rightarrow q \cdot \int_0^h dx = K \int_0^{\Delta T} dT \Rightarrow q \cdot h = K \Delta T \Rightarrow$
 $\Rightarrow q = \frac{K \Delta T}{h} = 30 \frac{\text{Pa}}{\text{m}^2}$.

~ 31 .
 $T_0 < T_1$ $K = \alpha T$ Q ?
 $q = K \frac{dT}{dx} = \frac{\alpha T dT}{dx} \Rightarrow Q = q S = \frac{\alpha T S dT}{dx} \Rightarrow$
 $\Rightarrow Q \cdot \int_0^L dx = \alpha S \int_{T_0}^{T_1} T dT \Rightarrow QL = \alpha S \left(\frac{T_1^2}{2} - \frac{T_0^2}{2} \right) \Rightarrow Q = \frac{\alpha S}{2L} (T_1^2 - T_0^2)$.

Н.У. 0.1:
 $V = 1 \text{ см}^3$, $d = 3.5 \cdot 10^{-8} \text{ м}$. Какое количество V ? $\chi \approx n \approx 1$, $\lambda = \frac{1}{\sqrt{2} n \pi d^2}$.
 За dt $\rightarrow dL = \langle v \rangle dt \Rightarrow \frac{\lambda}{\lambda \sqrt{2} dt} \text{ частиц}$ $\Rightarrow \lambda dt = \frac{\lambda \sqrt{2} dt}{\lambda} \cdot \frac{n}{2} \Rightarrow$
 $\Rightarrow \lambda = \frac{\sqrt{2} n^2 \pi d^2 \lambda \sqrt{2}}{2} = 2 \sqrt{\frac{\pi}{m}} \frac{p^2 d^2}{(n \pi)^2} = 0.6 \cdot 10^{-29} \frac{1}{\text{см}^3 \cdot \text{с}}$

№ 10.15. мен. выпл.
 Услов: $R = 10 \text{ см}$, $q = 100 \frac{\text{Вт}}{\text{см}^2}$, $T_0 = 373 \text{ К}$, $K = 400 \frac{\text{Вт}}{\text{м} \cdot \text{К}}$. $T(r)$ -? T_m -?
 Упр. ме гл. уапа: $\rho c \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) + q(r)$.
 Вектор стационарности $\Rightarrow \frac{\partial T}{\partial t} = 0 \Rightarrow -qr^2 dr = k d(r^2 \frac{\partial T}{\partial r}) \Rightarrow$
 $\Rightarrow -q \frac{r^3}{3} + \text{const} = k r^2 \frac{dT}{dr} \Rightarrow \frac{dT}{dr} = -\frac{q r}{3k} + \frac{\text{const}}{r^2}$
 Т.к. при $r=0 \rightarrow T \rightarrow +\infty \Rightarrow \text{const} = 0 \Rightarrow \int_{T_0}^T dT = \int_R^r -\frac{q r}{3k} dr \Rightarrow$
 $\Rightarrow T - T_0 = \frac{q}{6k} (R^2 - r^2)$; $T_3 = T_0 + \frac{q R^2}{6k} \approx 790 \text{ К}$.

№ 10.36.
 Услов: T_0, P_0 . $\chi = \chi(r)$. P -? $\chi = \frac{1}{3} \lambda \langle v \rangle n c_v$. $\lambda = \frac{1}{\sigma n} = \frac{kT}{\sigma P}$; $\langle v \rangle = \sqrt{\frac{3kT}{m}}$.
 $n = \frac{P}{kT} \Rightarrow \chi \propto \sqrt{T} \Rightarrow \chi = A \sqrt{T}$.
 $P_0 V = \frac{m}{M} RT \Rightarrow m = \frac{MP_0 V}{RT_0} = \frac{MP_0 SL}{RT_0} \Rightarrow dm = \frac{MPS}{RT(z)} dz \Rightarrow$
 $\Rightarrow m = \int_0^L \frac{MPS}{RT(z)} dz = \frac{MPS}{R} \int_0^L \frac{dz}{T(z)}$. $\dot{Q} = j_{\dot{Q}} S = -\chi S \frac{dT}{dz} \Rightarrow$
 $z = A \sqrt{T} S \frac{dT}{dz} \Rightarrow \dot{Q} dz = -SA \sqrt{T} dT \Rightarrow dz = \frac{-SA}{\dot{Q}} \sqrt{T} dT \Rightarrow$
 $\Rightarrow m = \frac{MPS}{R} \int_0^L \frac{dz}{T(z)} = \frac{MPS^2 A}{R \dot{Q}} \int_{T_0}^{T_1} \frac{\sqrt{T}}{T(z)} dT = \frac{A MPS^2}{R \dot{Q}} \int_{T_0}^{T_1} \frac{dT}{\sqrt{T}} = \frac{2 MPS^2 A}{R \dot{Q}} (\sqrt{T_1} - \sqrt{T_0})$
 $\Rightarrow \frac{MP_0 SL}{RT_0} = \frac{2 MPS^2 A}{R \dot{Q}} \sqrt{T_0} \Rightarrow P = \frac{P_0 L \dot{Q}}{T_0 S A \sqrt{2} \sqrt{T_0}}$.
 $\dot{Q} = -SA \sqrt{T} \frac{dT}{dz} = \text{const} \Rightarrow \int_0^L \dot{Q} dz = -SA \int_{T_0}^{T_1} \sqrt{T} dT \Rightarrow \frac{\dot{Q} L}{SA} = \frac{14}{3} T_0^{3/2} \Rightarrow$
 $\Rightarrow P = \frac{P_0}{T_0} \cdot \frac{1}{\sqrt{2}} \cdot \frac{14}{3} T_0^{3/2} = \frac{14}{3} P_0$.

№ 10.406.

 $t = 27^\circ \text{C}$, $\varphi = 50\%$,
 $L = 10 \text{ см}$, $l = 10^{-5} \text{ см}$,
 $P_H = 20 \text{ кПа}$ пр. ст. $T_{\text{н}} = ?$
 $j_x = -D \frac{\partial n}{\partial x} = -\frac{1}{3} \lambda \langle v \rangle \frac{n' - n''}{x}$; $P_n = n k T$;
 $P = \varphi P_n = n' k T$.

$$\mu - \mu^* = \frac{V P_H}{kT} - \frac{P_H}{kT} = \frac{P_H}{kT} (\psi - 1) = - \frac{P_H}{2kT} \Rightarrow j_x = \frac{1}{3} \lambda \langle v \rangle \frac{P_H}{2kT} \Rightarrow dN =$$

$$= dm \cdot \frac{N_A}{\mu} = \rho S dx \frac{N_A}{\mu} = j_x S dx \Rightarrow dx = \frac{\rho N_A}{\mu j_x} dx = \frac{\rho N_A \cdot 6kT}{\mu \lambda \langle v \rangle P_H} x dx \Rightarrow$$

$$\Rightarrow \tau = \frac{6 \rho R T}{\mu \lambda \langle v \rangle P_H} \int_{x/2}^L x dx = \frac{3}{4} \frac{\rho R T L^2}{\lambda \langle v \rangle P_H} \approx 227 \text{ с.м.}$$



$$T_1 = 330 \text{ K}; T_2 = 300 \text{ K};$$

$$K = 2.7 \cdot 10^{-2} \frac{\text{Вт}}{\text{К.м}}, R_2 = 2.7.$$

T10.

$$dS = \frac{\delta Q}{T}, q = k \pi \frac{dT}{dr} = -k 2\pi r L \frac{dT}{dr} \Rightarrow$$

$$\Rightarrow \dot{S} = \frac{-k 2\pi r}{T} \frac{dT}{dr} \Rightarrow$$

$$\Rightarrow \frac{\dot{S}}{L} \frac{dr}{r} = -k 2\pi \frac{dT}{T} \Rightarrow \frac{\dot{S}}{L} \ln \frac{R_2}{R_1} = -2\pi k \ln \frac{T_2}{T_1} \Rightarrow$$

$$\Rightarrow \frac{\dot{S}}{L} = \frac{2\pi k \ln \frac{T_2}{T_1}}{\ln \frac{R_2}{R_1}} \approx 1,63 \cdot 10^{-2} \frac{\text{Вт}}{\text{К.м.}}$$

1-ая раз; $V_2 = 2V_1$; $C = \frac{C_p}{2} = \text{const.}$

$\frac{z_2}{z_1} = ?$ $\frac{\lambda_2}{\lambda_1} = ?$

$$z = 4 \sqrt{\frac{\pi}{m k T}} p d^2; \lambda = \frac{\langle v \rangle}{z}; \langle v \rangle = \sqrt{\frac{8 k T}{\pi m}} \Rightarrow$$

$$\Rightarrow z \propto \frac{p}{\sqrt{T}} \propto \frac{1}{V}; \lambda \propto \frac{T}{p} \propto V \Rightarrow \frac{\lambda_2}{\lambda_1} = \frac{V_2}{V_1} = 2.$$

Процесс полиотропический: $C = \frac{C_p}{2} = \frac{\delta Q}{dT} = C_v + p \frac{dV}{dT} = C_v + \frac{RT}{V} \frac{dV}{dT} \Rightarrow$

$$\Rightarrow \left(\frac{5}{2} - \frac{3}{2}\right) R \frac{dT}{T} = R \frac{dV}{V} \Rightarrow -\frac{1}{4} \frac{dT}{T} = \frac{dV}{V} \Rightarrow \frac{1}{4} \ln \frac{T_2}{T_1} = \ln \frac{V_1}{V_2} \Rightarrow$$

$$\Rightarrow \frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^4 \Rightarrow T_2 = 2^{-4} T_1 \Rightarrow \frac{z_2}{z_1} = \sqrt{\frac{T_2}{T_1}} \frac{V_1}{V_2} = 2^{-2} \cdot \frac{1}{2} = \frac{1}{8}.$$

во 333а

$\alpha_0 = 0,14$; $E = 470 \frac{\text{к.Вт}}{\text{м.м.}}$; $W = 1 \text{ кВт}$; $D = 10^{-2} \frac{\text{м.м.}}{\text{с.}}$ $S = ?$

C_1 крибб $\alpha_0 = 0,14$; $C_2 = 2 \frac{\text{м.м.}}{\text{с.}}$; $h = 1 \text{ м.м.}$

температура (мм.): $\Phi = \frac{W}{E} \cdot N_A = -D \frac{dn}{dx} S = \left(\frac{\alpha_0 p_0}{kT} -$

$$-C_1 N_A) \cdot \frac{SD}{h} \Rightarrow S = \frac{Wh}{ED \left(\frac{\alpha_0 p_0}{RT} - C_2\right)} \approx 59 \text{ м.м.}^2.$$

N10.16.



$T(r) = ?$ процесс изотермический $\Rightarrow$ в ст. изотерм

$\rho C_V \frac{\partial T}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} (kr \frac{\partial T}{\partial r}) + q = 0$. Доп. $A = kr \frac{\partial T}{\partial r} \Rightarrow$
 $\Rightarrow dA = -q r dr \Rightarrow A = -q \frac{r^2}{2} + \text{const} = kr \frac{\partial T}{\partial r} \Rightarrow$
 $\Rightarrow \frac{\partial T}{\partial r} = \frac{-qr}{2k} + \frac{\text{const}}{r}$. При $r \rightarrow 0 \rightarrow \frac{\partial T}{\partial r} \rightarrow +\infty \Rightarrow \text{const} = 0$.
 $\Rightarrow \int dT = \frac{-q}{2k} \int r dr \Rightarrow T(r) - T_0 = \frac{-q}{2k} (\frac{r^2}{2} - \frac{R^2}{2}) \Rightarrow$
 $\Rightarrow T(r) = \frac{q}{4k} (R^2 - r^2) + T_0$. q - менно, безразмер.
 Заг. об'єм за зг. впрямку $\Rightarrow$ по закону Дре-Денуа:
 $q = I^2 \frac{R_{\text{середн}}}{V} = I^2 \frac{L}{S L} = \frac{I^2 \rho}{(\pi R^2)^2} \Rightarrow T(r) = T_0 + \frac{I^2 \rho}{4\pi^2 R^4 k} (R^2 - r^2)$

$\sqrt{10.243}$
 Канал логри. $\frac{\Phi_{\text{max}}}{\Phi_{\text{середн}}} - ?$ $\Phi = S j_{\text{кан}} = S \frac{n_{\text{max}} \langle v \rangle}{4}$.
 $R = 1 \text{ см}, \lambda = 10^{-5} \text{ м}$.

В беззвуче n логри роз-ми канал $\approx n_{\text{max}} \Rightarrow$
 $\Rightarrow \Phi_{\text{середн}} = S j_{\text{середн}} = -D \frac{dn}{dr} S \approx D \frac{n_{\text{max}}}{R} S = \frac{1}{3} \lambda \langle v \rangle \frac{n_{\text{max}} S}{R} \Rightarrow$
 $\Rightarrow \frac{\Phi_{\text{max}}}{\Phi_{\text{середн}}} = \frac{3 R}{4 \lambda} = 0,75 \cdot 10^5$

$\sqrt{10.25}$
 Канал логри & беззвуче:
 $a = 0,2 \text{ м}; t_1 = 0^\circ \text{C}; t_2 = -10^\circ \text{C}; \tau = ?$
 $q = 335 \frac{\text{Вт}}{\text{м}^2}; k = 0,022 \frac{\text{Вт}}{\text{К} \cdot \text{м}}; \rho \approx 7800$

Менно, беззвуче при замержанні асн: $dQ =$
 $= q dm = q \rho \cdot 4\pi r^2 dr$. ЗСД: $q \rho \cdot 4\pi r^2 dr = 4\pi x^2 k \frac{\partial T}{\partial x} dt_{\text{ф}}$
 $\Rightarrow \int \rho q r^2 dr \int \frac{dx}{x^2} = k \int dT dt_{\text{ф}} \Rightarrow k (T_1 - T_2) dt_{\text{ф}} = \rho q r^2 dr (\frac{1}{a} - \frac{1}{r})$
 $\Rightarrow \frac{\rho q}{4k(T_1 - T_2)} (\int_0^a r^2 dr - a \int_0^a r dr) = \int_0^a dt_{\text{ф}} \Rightarrow \tau = \frac{\rho q a^2}{6k(T_1 - T_2)} \approx 10 \text{ с}$

6

$\sqrt{32}$
 Аппарат: $R \sim 10 \text{ мкм}, \eta \sim 2 \cdot 10^{-5} \text{ Па} \cdot \text{с}$. Закон Ампера: $E_p = 6\pi\eta r \dot{\gamma}$; $\langle E \rangle = \frac{3}{2} k T_0 \dot{\gamma}$

$$\Rightarrow \Delta x \sim \sqrt{\frac{3kT}{2.6 \cdot 10^{21} \cdot 10}} \cdot D = \frac{1}{3} \langle v \rangle \lambda \Rightarrow$$

$$\Rightarrow D \sim \Delta x \sqrt{N} = \frac{kT}{4 \pi \eta r} \sim 10^{-12} \text{ м}^2/\text{с}$$

33.

MCN & longwave: $\Delta x \sim 100 \text{ нм}$, $\lambda \sim 10^{-5} \text{ м}$, $\alpha = ?$

$$D \sim \frac{\Delta x^2}{t} \Rightarrow t \sim \frac{\Delta x^2}{D}; D = \frac{1}{3} \lambda \langle v \rangle = \left(\frac{\mu_{MCN}}{\mu_{long}} \right)$$

$$= \frac{1}{3} \lambda \sqrt{\frac{8RT}{\pi \mu}} \Rightarrow t \sim \frac{\Delta x^2}{D} = 3 \frac{\Delta x^2}{\lambda} \sqrt{\frac{\pi \mu}{8RT}} \Rightarrow t \sim 100 \text{ с}$$

34.

T_1, T_2

$d \ll \lambda, \frac{P_2}{P_1} = ?$ $\text{Получено по тем же: } S \frac{n_1 v_1}{4} = S \frac{n_2 v_2}{4} \Rightarrow$

$$\Rightarrow \frac{P_1 v_1}{T_1} = \frac{P_2 v_2}{T_2}, v \propto \sqrt{T} \Rightarrow \frac{P_1}{P_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{P_2}{P_1} = \sqrt{2}$$

35.

$\lambda > d = 1 \text{ см (for. hor)}$ $D = ?$ $\lambda \text{ оп. разнесены по углам}$

$$D = \frac{1}{3} d \langle v \rangle = \frac{1}{3} d \sqrt{\frac{8RT}{\pi \mu}} \approx 1,6 \text{ м}^2/\text{с}$$

T 12.

данные: $T = 10^\circ \text{C}$, $\lambda = 0,5 \text{ м}$, $t = 12$, $\langle r^2 \rangle = ?$ $D = ?$

$$\Delta \bar{r} = \sum_{i=1}^N \bar{r}_i; N = \frac{t}{\tau} \Rightarrow \Delta \bar{r}^2 =$$

$$= \sum_{i=1}^N \bar{r}_i^2 + 2 \sum_{i < j} (\bar{r}_i, \bar{r}_j) = \lambda^2 N \Rightarrow \sqrt{\langle \Delta r^2 \rangle} = \lambda \sqrt{\frac{t}{\tau}} = 30 \text{ м}$$

Для малых $R \ll r \Rightarrow \langle R^2 \rangle \ll \langle r^2 \rangle$ $\langle r^2 \rangle = 4Dt \Rightarrow D = \frac{\langle r^2 \rangle}{4t} = 225 \text{ м}^2/\text{с}$

10.92.

$m = 10^{-40} \text{ г}$, $h = 1 \text{ м}$, $T = 300 \text{ К}$, $\rho = 0,9 \text{ г/см}^3$, $\eta = 1,8 \cdot 10^{-4} \text{ П}$ $\langle r^2 \rangle = ?$ $t = \frac{h}{u}$; $u = \sqrt{F} \Rightarrow t = \frac{h}{\sqrt{mg}}$

$$D = \frac{kT}{mg} \Rightarrow t = \frac{h}{\sqrt{mg}}$$

$$= 4Dt = \frac{4kTh}{mg} \approx 1,7 \cdot 10^{-8} \text{ м}^2 \Rightarrow \sqrt{\langle r^2 \rangle} = 1,3 \cdot 10^{-4} \text{ м}$$

Функция плотности ρ по Стокса: радиус капли $a = \sqrt[3]{\frac{3m}{4\pi\rho}} = 1,36 \cdot 10^{-5} \text{ см} \Rightarrow \lambda \approx 10^{-5} \text{ м}$

$$Re = \frac{mgR}{6\eta r^2} \approx 0,16 < 1$$

№ 10.68 (Безкоординат)

$V=100\text{ л}, r=2\text{ см}, L=1\text{ м},$
 $p_1=10^5\text{ Па}, p_2=10^4\text{ Па}, \eta=1,8 \cdot 10^{-3}\text{ Па}\cdot\text{с}$
 $t=?$ *Множество: $\dot{m} = 0 \frac{p_1 - p_2}{r^2 L \eta}$*
 $\text{Реш: } \rho = \frac{pM}{RT} = \frac{p_1 + p_2}{2} \frac{M}{RT} \Rightarrow \dot{m} = \frac{dm}{dt} = \frac{4\pi r^4 \eta (p_1^2 - p_2^2)}{16\eta L RT}$

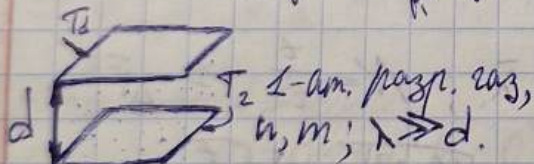
Начи углубление $\Rightarrow p_2 = 0, p_1 = p$
 $m = \frac{pV}{RT} \Rightarrow \frac{dm}{dt} = \frac{V}{RT} \frac{dp}{dt}$
 $= \frac{4\pi r^4 \eta p^2}{16\eta L RT} \Rightarrow \frac{16\eta LV}{4\pi r^4} \int_{p_2}^{p_1} \frac{dp}{p^2} = \int_0^t dt \Rightarrow t = \frac{16\eta LV}{4\pi r^4} \left(\frac{1}{p_1} - \frac{1}{p_2} \right) \approx 430$

№ 10.69 (Разр. раз)

Все условия; $p_1=10^5\text{ Па}, p_2=10^4\text{ Па}$
 $\Phi = -D \frac{dn}{dx} 4\pi r^2 =$
 $= \frac{1}{3} \lambda \langle v \rangle 4\pi r^2 \frac{n_1 - n_2}{L} \Rightarrow \text{т.к. } \lambda = 2r, n = \frac{p}{kT} \Rightarrow \Phi = \frac{2}{3} 4\pi r^3 \langle v \rangle \frac{p_1 - p_2}{L kT}$

$\frac{dM}{dt} = m \Phi \frac{M}{M} = \frac{4}{3} \frac{r^3}{L} \sqrt{\frac{2\pi M}{RT}} \Delta p$ (в-е из Кнудсена)
 $M = \frac{pV}{RT} \Rightarrow \frac{dM}{dt} = - \frac{V}{RT} \frac{dp}{dt} = - \frac{4}{3} \sqrt{\frac{2\pi M}{RT}} \frac{r^3}{L} p$ *В газе $\Delta p = p$*

$t = \int_0^t dt =$
 $= - \frac{V}{RT} \frac{3}{4} \sqrt{\frac{RT}{2\pi M}} \frac{L}{r^3} \int_{p_1}^{p_2} \frac{dp}{p} = \frac{3}{4} \frac{LV}{r^3} \sqrt{\frac{M}{2\pi RT}} \ln \frac{p_1}{p_2} \approx 88\text{ с}$



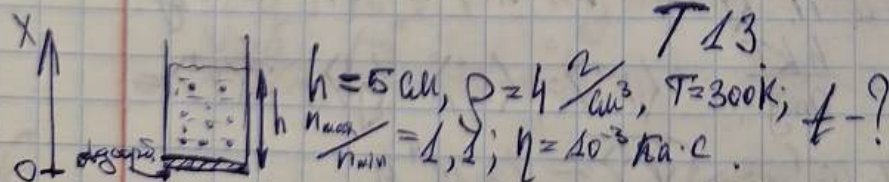
№ 10.120.

$q \left[\frac{\text{Дж}}{\text{м}^2 \cdot \text{с}} \right] = ?$ $n = n_1 + n_2$

Помощь равна $\Rightarrow n_1 \sqrt{T_1} = n_2 \sqrt{T_2} \Rightarrow n_1 = \frac{\sqrt{T_2}}{\sqrt{T_1} + \sqrt{T_2}} n$; $n_2 = \frac{\sqrt{T_1}}{\sqrt{T_1} + \sqrt{T_2}} n$

$\epsilon = 2kT \Rightarrow q = \frac{1}{4} n_1 \langle v_1 \rangle 2kT_1 - \frac{1}{4} n_2 \langle v_2 \rangle 2kT_2 = \frac{nk}{2} \sqrt{\frac{8k}{\pi m}}$

$\cdot \frac{\sqrt{T_2 T_1 T_1} - \sqrt{T_2 T_1 T_2}}{\sqrt{T_1} + \sqrt{T_2}} = n \sqrt{\frac{2k^3}{\pi m}} \sqrt{T_1 T_2} (\sqrt{T_1} - \sqrt{T_2})$



Т 13

Дрейф

$n(x) = n_0 \exp\left(-\frac{mgx}{kT}\right) \Rightarrow m_{\text{эф}} = \frac{kT}{gh} \ln \frac{n_0}{n(h)} = \frac{kT}{gh} \ln \frac{n_{\text{max}}}{n_{\text{min}}} \approx 7,9 \cdot 10^{-22}\text{ кг}$

$$V = \frac{m_{\text{max}}}{\rho - \rho_0} = \frac{4}{3} \pi r^3 \Rightarrow r = \sqrt[3]{\frac{3}{4\pi} \frac{m_{\text{max}}}{\rho - \rho_0}} = 4 \cdot 10^{-9} \text{ м. } t_{\text{gr}} = \frac{h}{\eta};$$

$$\eta = BF; B = \frac{1}{6\pi\eta r}; F = m_{\text{max}} g \Rightarrow t_{\text{gr}} = \frac{6h\eta r}{m_{\text{max}} g} \approx 4,7 \cdot 10^8 \text{ с} \sim 15 \text{ лет.}$$

Диффузия: $t_{\text{diff}} = \frac{h^2}{2D}$. По закону Фика: $D = BkT \Rightarrow$

$$\Rightarrow t_{\text{diff}} = \frac{h^2}{2BkT} = \frac{3h^2 \eta r}{kT} \approx 2,3 \cdot 10^7 \text{ с} \sim 9 \text{ мес.} \Rightarrow$$

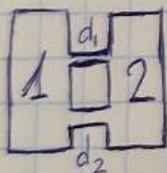
Пластина будет оседать за счёт диффузии за ~ 9 мес.

Пример задачи: $\tau \sim 10^{-10} \text{ с}$; $\kappa \sim 1 \frac{\text{Вт}}{\text{м} \cdot \text{К}}$; $c \sim 10^6 \frac{\text{Дж}}{\text{м}^3 \cdot \text{К}}$. h - ?

$$\rho C \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(\kappa \frac{\partial T}{\partial x} \right) \Rightarrow \rho C \frac{\partial T}{\partial t} = \kappa \frac{\partial^2 T}{\partial x^2} \Rightarrow \frac{\partial T}{\partial t} \sim \frac{\Delta T}{\tau}; \frac{\partial^2 T}{\partial x^2} \sim \frac{\Delta T}{h^2} \Rightarrow \frac{\Delta T}{\tau} \sim \frac{\kappa}{C} \frac{\Delta T}{h^2} \Rightarrow h \sim \sqrt{\frac{\kappa \tau}{C}} \approx 3 \text{ м.}$$

Ar-40: $\lambda = 10^{-5} \text{ см}$, $E = 300 \frac{\text{эВ}}{\text{см}}$, $T \sim 300 \text{ К}$. $v_{\text{gr}} = ?$ $v_{\text{gr}} = BF =$

$$= eEB = eE \frac{D}{kT} = \frac{1}{3} \frac{\langle v \rangle \lambda e E}{kT} = \frac{\sqrt{8RT}}{\sqrt{\pi M}} \frac{\lambda e E}{3kT} \sim 15 \text{ м/с.}$$



$V_1 = V_2 = V$; $T_1 = 800 \text{ К}$; $T_2 = 200 \text{ К}$; m - ? ① Широкая трубка

$d_1 \ll \lambda$; $d_2 \gg \lambda$; $M_1 + M_2 = M$.

Открыта $\Rightarrow p_1 = p_2 \Rightarrow n_1 T_1 = n_2 T_2 \Rightarrow \frac{n_1}{n_2} = \frac{T_2}{T_1}$.

$$M_i = \rho_i V = m_0 n_i V \Rightarrow \frac{M_1}{M_2} = \frac{n_1}{n_2} = \frac{T_2}{T_1} \Rightarrow M_1 = \frac{M}{1 + \frac{T_1}{T_2}} = \frac{M T_2}{T_2 + T_1} = \frac{M}{5}.$$

② Широкая трубка перекрыта $\Rightarrow$ равенство потоков: $j_{1 \rightarrow 2} = j_{2 \rightarrow 1}$

$$\Leftrightarrow \frac{1}{4} n_1' \langle v_1 \rangle = \frac{1}{4} n_2' \langle v_2 \rangle \Rightarrow n_1' \sqrt{T_1} = n_2' \sqrt{T_2} \Rightarrow \frac{n_1'}{n_2'} = \frac{n_1}{n_2} = \sqrt{\frac{T_2}{T_1}} \Rightarrow$$

$$\Rightarrow M_1' = \frac{M}{1 + \sqrt{\frac{T_1}{T_2}}} = \frac{M \sqrt{T_2}}{\sqrt{T_2} + \sqrt{T_1}} = \frac{M}{3} > M_1 = \frac{M}{5} \Rightarrow \text{при пере-}$$

крытии газ будет перетекать из 2-го сосуда в 1-ый.

$$m = M_1' - M_1 = M \left(\frac{1}{3} - \frac{1}{5} \right) = \frac{2}{15} M.$$

Макс. работа от изм. изм. бодов и 1 кр кар.? $\Delta S = C \ln \frac{T}{T_1} +$
 $+ C \ln \frac{T}{T_2} = 0 \Rightarrow T = \sqrt{T_1 T_2} \Rightarrow A = C(T_2 - T) - C(T - T_1).$